

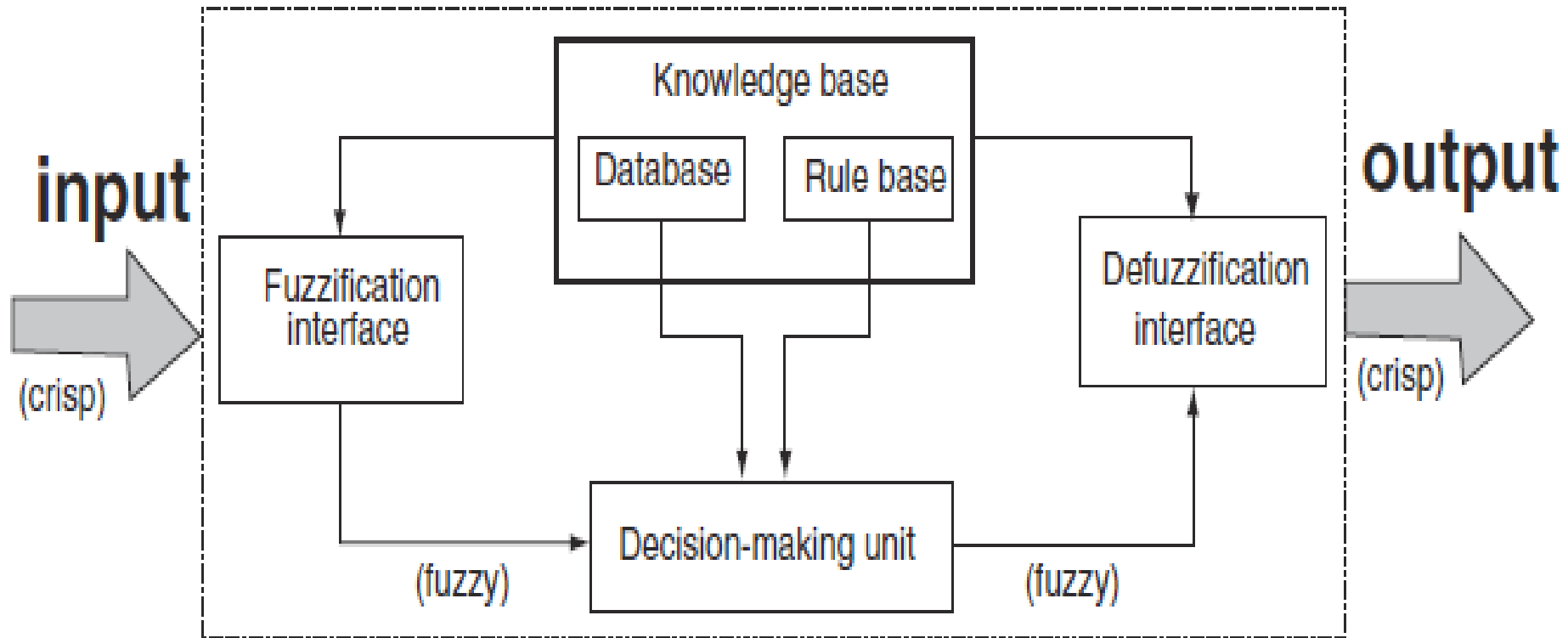
MODULE 4

FUZZY INFERENCE SYSTEMS

FUZZY INFERENCE SYSTEMS (FIS)

- Fuzzy rule based systems, fuzzy models, and fuzzy expert systems are also known as fuzzy inference systems.
- The key unit of a fuzzy logic system is FIS.
- The primary work of this system is decision-making.
- FIS uses “IF...THEN” rules along with connectors “OR” or “AND” for making necessary decision rules.
- The input to FIS may be fuzzy or crisp, but the output from FIS is always a fuzzy set.
- When FIS is used as a controller, it is necessary to have crisp output.
- Hence, there should be a defuzzification unit for converting fuzzy variables into crisp variables along FIS.

BLOCK DIAGRAM OF FIS



Construction and Working of Inference System

Fuzzy inference system consists of a fuzzification interface, a rule base, a database, a decision-making unit, and finally a Defuzzification interface

- a **rule base** containing a number of fuzzy IF–THEN rules;
- a **database** which defines the membership functions of the fuzzy sets used in the fuzzy rules;
- a **decision-making unit** which performs the inference operations on the rules;
- a **fuzzification interface** which transforms the crisp inputs into degrees of match with linguistic values; and
- a **Defuzzification interface** which transforms the fuzzy results of the inference into a crisp output.

WORKING OF FIS

- **The crisp input is converted in to fuzzy by using fuzzification method.**
- **After fuzzification the rule base is formed.**
- **The rule base and the database are jointly referred to as the knowledge base.**
- **Defuzzification is used to convert fuzzy value to the real world value which is the output.**

The steps of fuzzy reasoning (inference operations upon fuzzy IF–THEN rules) performed by FISs are:

1. Compare the input variables with the membership functions on the antecedent part to obtain the membership values of each linguistic label (this step is often called **fuzzification**.)
2. Combine (through a specific t-norm operator, usually multiplication or min) the membership values on the premise part to get firing strength (weight) of each rule.
3. Generate the qualified consequents (either fuzzy or crisp) for each rule depending on the firing strength.
4. Aggregate the qualified consequents to produce a crisp output. (This step is called **Defuzzification**.)

TYPES OF FIS

There are two types of Fuzzy Inference Systems:

- Mamdani FIS(1975)
- Sugeno FIS(1985)

MAMDANI FUZZY INFERENCE SYSTEMS (FIS)

- **Mamdani's fuzzy inference method is the most commonly seen fuzzy methodology.**
- **Mamdani's method was among the first control systems built using/fuzzy set theory.**
- **It was proposed by Mamdani (1975) as an attempt to control a steam engine and boiler combination by synthesizing a set of linguistic control rules obtained from experienced human operators.**
- **Mamdani's effort was based on Zadeh's (1973) paper on fuzzy algorithms for complex systems and decision processes.**

MAMDANI FUZZY INFERENCE SYSTEMS (FIS)

- Mamdani type inference expects the output membership functions to be fuzzy sets.
- After the aggregation process, there is a fuzzy set for each output variable that need Defuzzification.
- It is possible, and in many cases much more efficient, to use a single spike as the output membership function rather than a distributed fuzzy set.
- This is sometimes known as a singleton output membership function, and it can be thought of as a pre - defuzzified fuzzy set.
- It enhances the efficiency of the Defuzzification process because it greatly simplifies the computation required by the more general Mamdani method, which finds the centroid of a two-dimensional function.

MAMDANI FUZZY INFERENCE SYSTEMS (FIS)

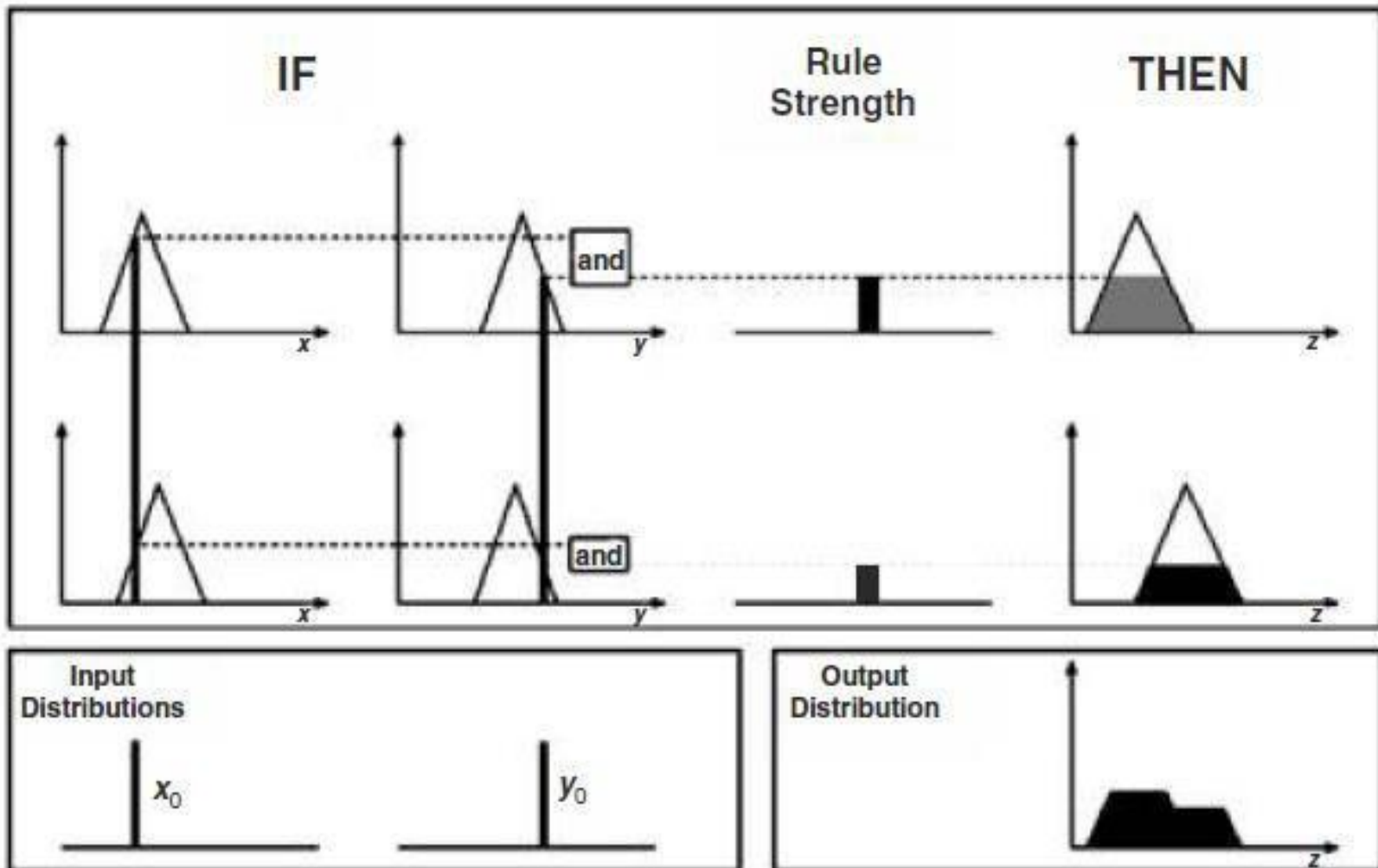


Fig. 6.2. A two input, two rule Mamdani FIS with crisp inputs

MAMDANI FUZZY INFERENCE SYSTEMS (FIS)

To compute the output of this FIS given the inputs, six steps has to be followed:

- 1. Determining a set of fuzzy rules**
- 2. Fuzzifying the inputs using the input membership functions**
- 3. Combining the fuzzified inputs according to the fuzzy rules to establish a rule strength**
- 4. Finding the consequence of the rule by combining the rule strength and the output membership function**
- 5. Combining the consequences to get an output distribution**
- 6. Defuzzifying the output distribution (this step is only if a crisp output (class) is needed).**

Creating Fuzzy Rules

- Fuzzy rules are a collection of linguistic statements that describe how the FIS should make a decision regarding classifying an input or controlling an output.
- Fuzzy rules are always written in the following form:
if (input 1 is membership function 1) and/or (input 2 is membership function 2) and/or. . . then (output n is output membership function n).
- For example:
 - if temperature is high and humidity is high then room is hot.
 - There would have to be membership functions that define high temperature (input 1), high humidity (input 2), and a hot room (output 1).
 - This process of taking an input such as temperature and processing it through a membership function to determine “high” temperature is called fuzzification and is discussed in section, “Fuzzification.”
 - Also, “AND”/“OR” in the fuzzy rule should be defined.
 - This is called fuzzy combination

Fuzzification

- The purpose of fuzzification is to map the inputs from a set of sensors (or features of those sensors such as amplitude or spectrum) to values from 0 to 1 using a set of input membership functions.
- In the example shown in Fig. 6.2, there are two inputs, x_0 and y_0 shown at the lower left corner.
- These inputs are mapped into fuzzy numbers by drawing a line up from the inputs to the input membership functions above and marking the intersection point.
- These input membership functions, as discussed previously, can represent fuzzy concepts such as “large” or “small,” “old” or “young,” “hot” or “cold,” etc.
- When choosing the input membership functions, the definition of large and small may be different for each input.

Consequence

The consequence of a fuzzy rule is computed using two steps:

1. Computing the rule strength by combining the fuzzified inputs using the fuzzy combination process discussed in previous section.

This is shown in Fig. 6.2.

In this example, the fuzzy “AND” is used to combine the membership functions to compute the rule strength.

2. Clipping the output membership function at the rule strength.

Combining Outputs into an Output Distribution

- The outputs of all of the fuzzy rules must now be combined to obtain one fuzzy output distribution.
- This is usually, but not always, done by using the fuzzy “OR.”
- The output membership functions on the right-hand side of the figure are combined using the fuzzy OR to obtain the output distribution shown on the lower right corner of the Fig. 6.2.

Defuzzification of Output Distribution

- In many instances, it is desired to come up with a single crisp output from an FIS.
- For example, if one was trying to classify a letter drawn by hand on a drawing tablet, ultimately the FIS would have to come up with a crisp number to tell the computer which letter was drawn.
- This crisp number is obtained in a process known as Defuzzification.

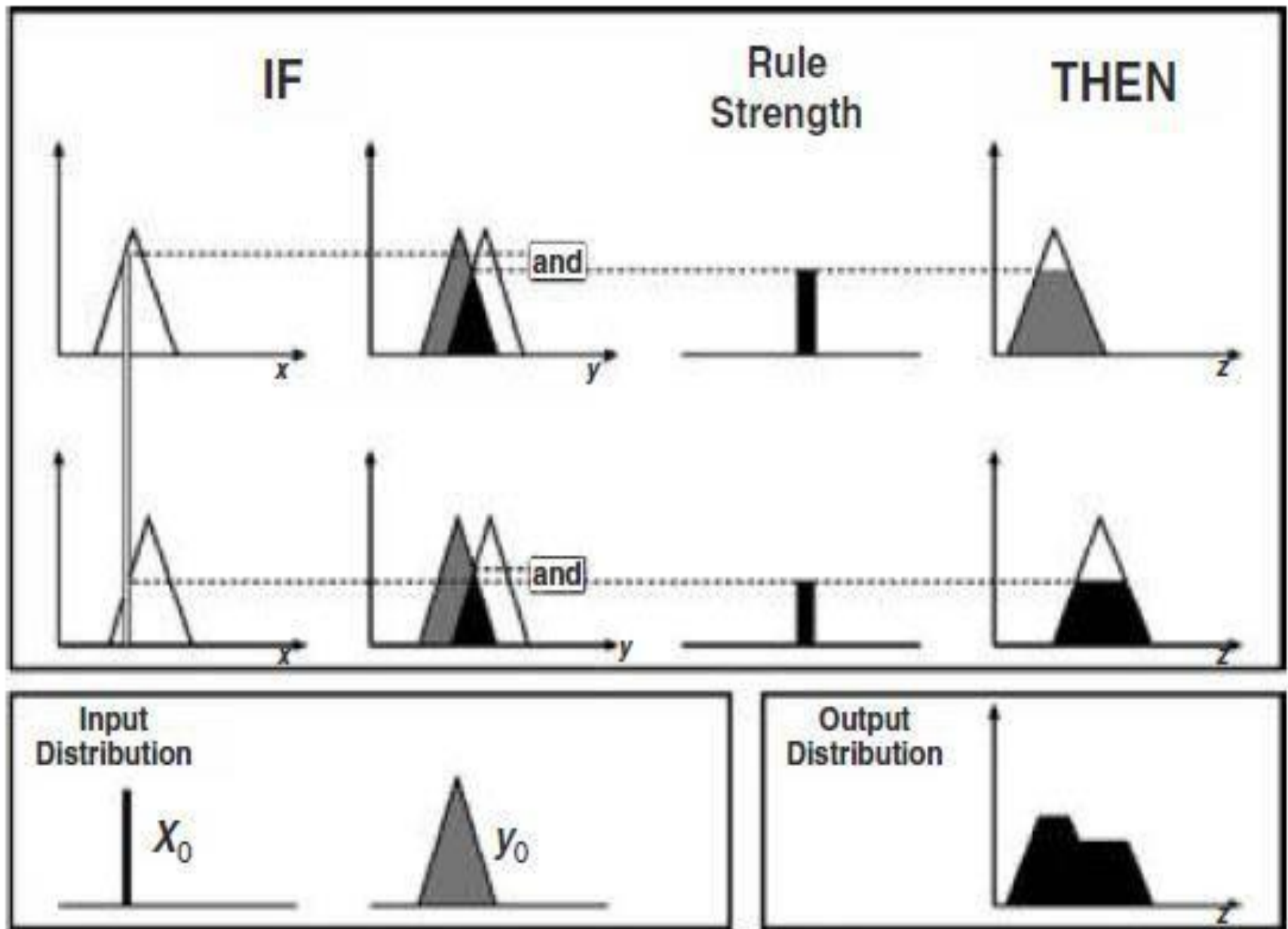


Fig. 6.5. A two input, two rule Mamdani FIS with a fuzzy input

MAMDANI FUZZY INFERENCE SYSTEMS (FIS)

- Fuzzify input variables:
 - Determine membership values.
- Evaluate rules:
 - Based on membership values of (composite) antecedents.
- Aggregate rule outputs:
 - Unify all membership values for the output from all rules.
- Defuzzify the output:
 - COG: Center of gravity (approx. by summation).

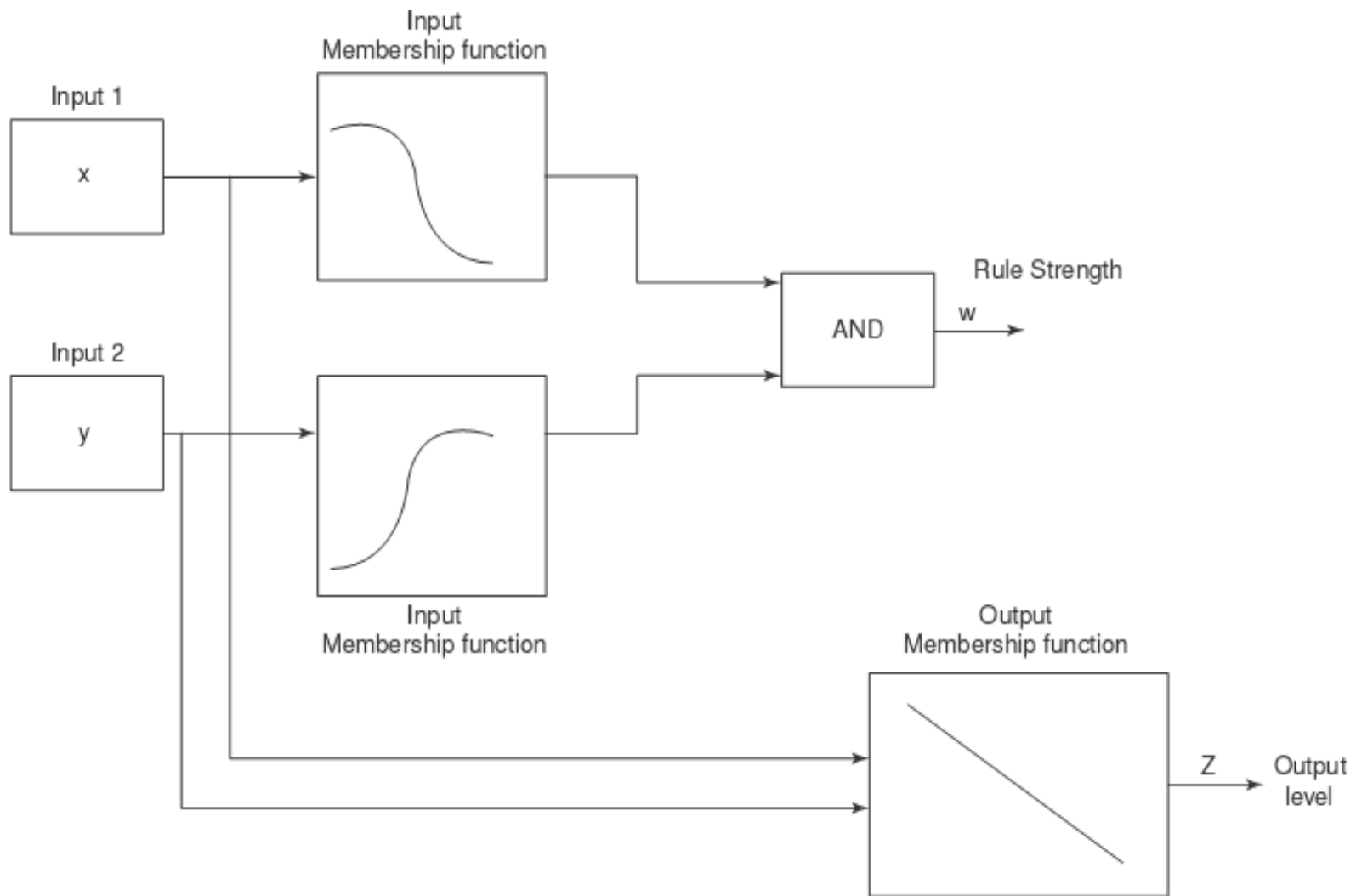
SUGENO FUZZY INFERENCE SYSTEMS (FIS)

The main steps of the fuzzy inference process namely,

1. fuzzifying the inputs and
2. applying the fuzzy operator are exactly the same as in MAMDANI FIS.

The main difference between Mamdani's and Sugeno's methods is that Sugeno output membership functions are either linear or constant.

SUGENO FIS



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Sugeno fuzzy model

- Sugeno fuzzy model is also known as Sugeno–Takagi model.
- A typical fuzzy rule in a Sugeno fuzzy model has the format

IF x is A and y is B THEN $z = f(x, y)$,

- where A, B are fuzzy sets in the antecedent;
- $Z = f(x, y)$ is a crisp function in the consequent

Sugeno fuzzy model

- Usually $f(x, y)$ is a polynomial in the input variables x and y , but it can be any other functions that can appropriately describe the output of the system within the fuzzy region specified by the antecedent of the rule.
- When $f(x, y)$ is a first-order polynomial, we have the first-order Sugeno fuzzy model.
- When f is a constant, we then have the zero-order Sugeno fuzzy model, which can be viewed either as a special case of the Mamdani FIS
- or a special case of Tsukamoto's fuzzy model where each rule's consequent is specified by a membership function of a step function centered at the constant.

Sugeno fuzzy model

- The first two parts of the fuzzy inference process, fuzzifying the inputs and applying the fuzzy operator, are exactly the same.
- The main difference between Mamdani and Sugeno is that the Sugeno output membership functions are either linear or constant.
- A typical rule in a Sugeno fuzzy model has the form

IF Input 1 = x AND Input 2 = y , THEN Output is $z = ax + by + c$.

Sugeno fuzzy model

- For a zero-order Sugeno model, the output level z is a constant ($a = b = 0$).
- The output level z_i of each rule is weighted by the firing strength w_i of the rule.
- For example, for an AND rule with Input 1 = x and Input 2 = y , the firing strength is

$$w_i = \text{AndMethod}(F_1(x), F_2(y)),$$

where $F_{1,2}(\cdot)$ are the membership functions for Inputs 1 and 2. The final output of the system is the weighted average of all rule outputs, computed as

$$\text{Final output} = \frac{\sum_{i=1}^N w_i z_i}{\sum_{i=1}^N w_i}.$$

Comparison Between Sugeno and Mamdani Method

- The main difference between Mamdani and Sugeno is that the Sugeno output membership functions are either linear or constant.
- Also the difference lies in the consequents of their fuzzy rules, and thus their aggregation and Defuzzification procedures differ suitably
- In Sugeno method a large number of fuzzy rules must be employed to approximate periodic or highly oscillatory functions.
- The minimal configuration of the TS fuzzy systems can be reduced and becomes smaller than that of the Mamdani fuzzy systems if nontrapezoidal or nontriangular input fuzzy sets are used.
- Far fewer mathematical results exist for TS fuzzy controllers than do for Mamdani fuzzy controllers, notably those on TS fuzzy control system stability.
- Mamdani is easy to form compared to Sugeno method.

Advantages of the Sugeno Method

- ✓ It is computationally efficient.
- ✓ It works well with linear techniques (e.g., PID control).
- ✓ It works well with optimization and adaptive techniques.
- ✓ It has guaranteed continuity of the output surface.
- ✓ It is well suited to mathematical analysis.

Advantages of the Mamdani Method

- **It is intuitive.**
- **It has widespread acceptance.**
- **It is well suited to human input.**

SUMMARY

- Advantages of fuzzy logic
 - Allows the use of vague linguistic terms in the rules.

- Disadvantages of fuzzy logic
 - Difficult to estimate membership function
 - There are many ways of interpreting fuzzy rules, combining the outputs of several fuzzy rules and de-fuzzifying the output.

- Single input single output Mamdani fuzzy model with 3 rules:
- **If X is small then Y is small $\rightarrow$ R1**
- **If X is medium then Y is medium $\rightarrow$ R2**
- **Is X is large then Y is large $\rightarrow$ R3**

$X = \text{input} \in [-10, 10]$ $Y = \text{output} \in [0, 10]$ Using centroid Defuzzification

Two input single-output Mamdani fuzzy model with 4 rules:

If X is small & Y is small then Z is negative large

If X is small & Y is large then Z is negative small

If X is large & Y is small then Z is positive small

If X is large & Y is large then Z is positive large

$X = [-5, 5]; Y = [-5, 5]; Z = [-5, 5]$

Single output-input Sugeno fuzzy model with three rules

If X is small then $Y = 0.1X + 6.4$

If X is medium then $Y = -0.5X + 4$

If X is large then $Y = X - 2$

**If “small”, “medium” & “large” are nonfuzzy sets
then**

**the overall input-output curve is a piece wise
linear**

Two-input single output fuzzy model with 4 rules

R1: if X is small & Y is small then $z = -x + y + 1$

R2: if X is small & Y is large then $z = -y + 3$

R3: if X is large & Y is small then $z = -x + 3$

R4: if X is large & Y is large then $z = x + y + 2$

Lecture 5

Fuzzy expert systems: Fuzzy inference

- **Mamdani fuzzy inference**
- **Sugeno fuzzy inference**
- **Case study**
- **Summary**

Fuzzy inference

The most commonly used fuzzy inference technique is the so-called Mamdani method. In 1975, Professor **Ebrahim Mamdani** of London University built one of the first fuzzy systems to control a steam engine and boiler combination. He applied a set of fuzzy rules supplied by experienced human operators.

Mamdani fuzzy inference

- The Mamdani-style fuzzy inference process is performed in four steps:
 - fuzzification of the input variables,
 - rule evaluation;
 - aggregation of the rule outputs, and finally
 - defuzzification.

We examine a simple two-input one-output problem that includes three rules:

Rule: 1

IF x is $A3$
OR y is $B1$
THEN z is $C1$

Rule: 2

IF x is $A2$
AND y is $B2$
THEN z is $C2$

Rule: 3

IF x is $A1$
THEN z is $C3$

Rule: 1

IF *project_funding* is *adequate*
OR *project_staffing* is *small*
THEN *risk* is *low*

Rule: 2

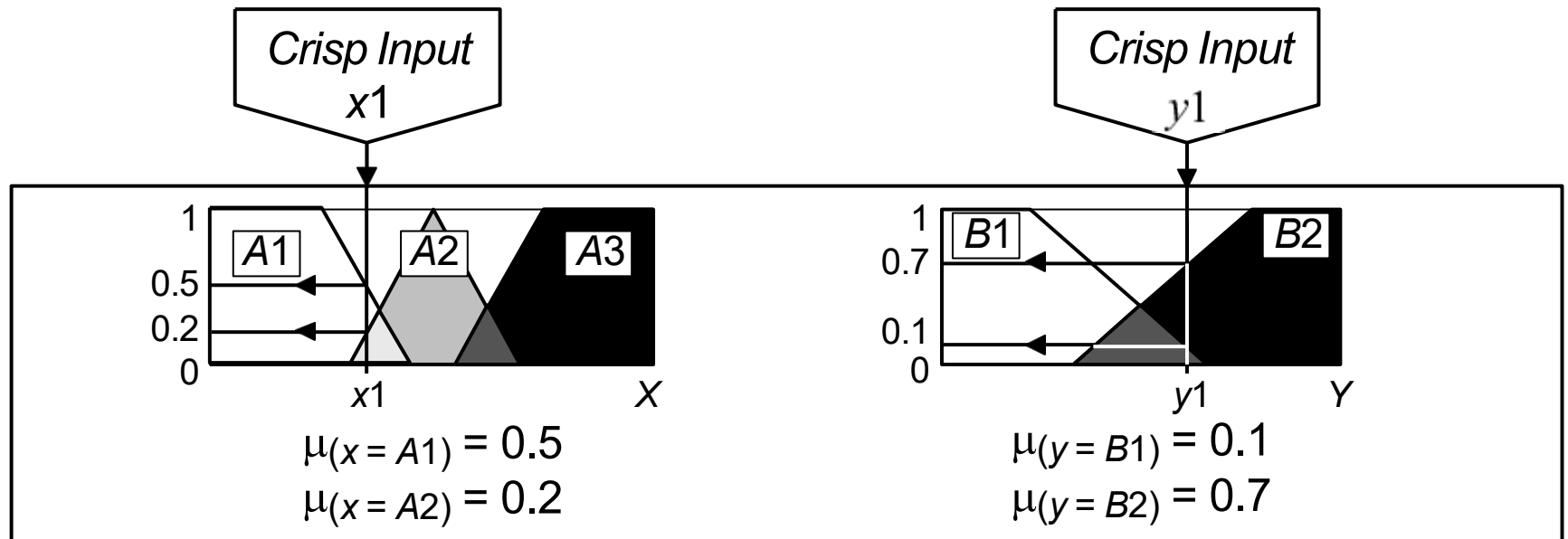
IF *project_funding* is *marginal*
AND *project_staffing* is *large*
THEN *risk* is *normal*

Rule: 3

IF *project_funding* is *inadequate*
THEN *risk* is *high*

Step 1: Fuzzification

The first step is to take the crisp inputs, x_1 and y_1 (*project funding* and *project staffing*), and determine the degree to which these inputs belong to each of the appropriate fuzzy sets.



Step 2: Rule Evaluation

The second step is to take the fuzzified inputs, $\mu_{(x=A1)} = 0.5$, $\mu_{(x=A2)} = 0.2$, $\mu_{(y=B1)} = 0.1$ and $\mu_{(y=B2)} = 0.7$, and apply them to the antecedents of the fuzzy rules. If a given fuzzy rule has multiple antecedents, the fuzzy operator (AND or OR) is used to obtain a single number that represents the result of the antecedent evaluation. This number (the truth value) is then applied to the consequent membership function.

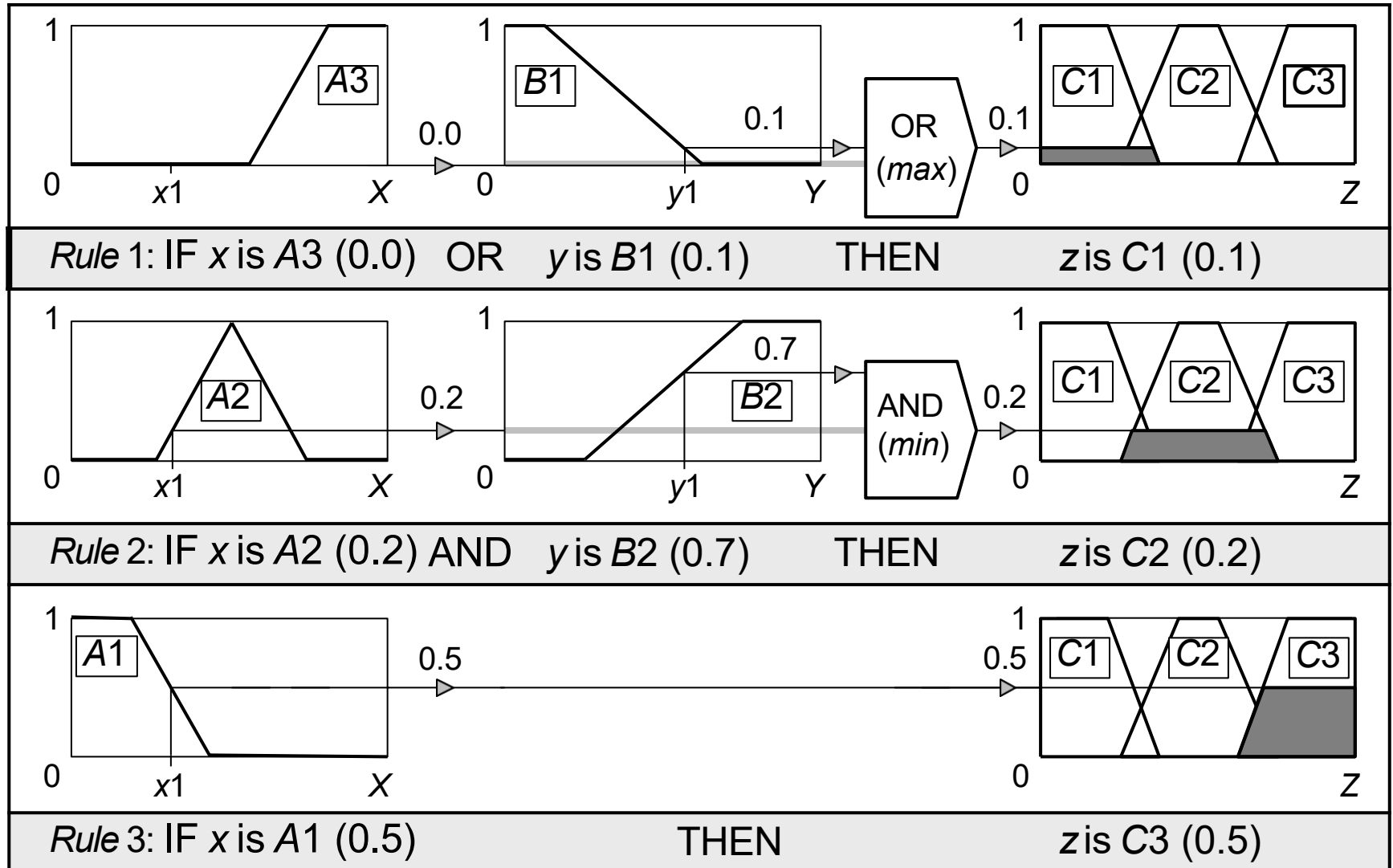
To evaluate the disjunction of the rule antecedents, we use the **OR fuzzy operation**. Typically, fuzzy expert systems make use of the classical fuzzy operation **union**:

$$\mu_{A \cup B}(x) = \max [\mu_A(x), \mu_B(x)]$$

Similarly, in order to evaluate the conjunction of the rule antecedents, we apply the **AND fuzzy operation intersection**:

$$\mu_{A \cap B}(x) = \min [\mu_A(x), \mu_B(x)]$$

Mamdani-style rule evaluation



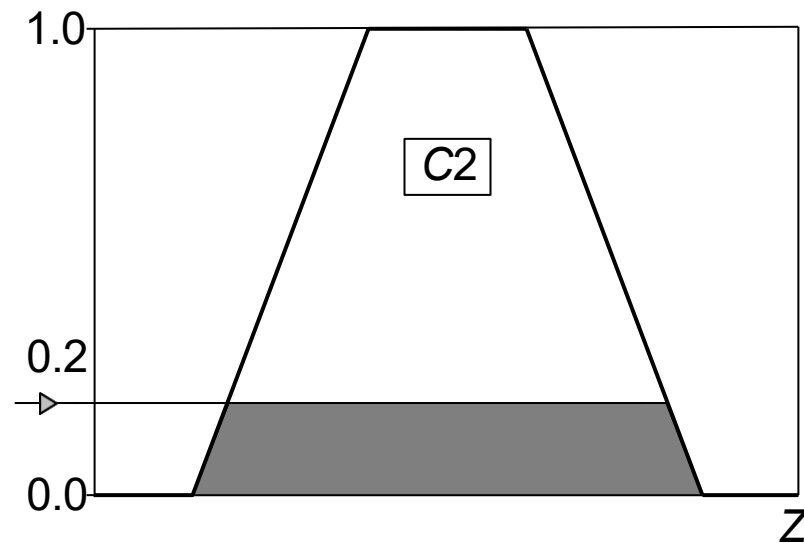
Now the result of the antecedent evaluation can be applied to the membership function of the consequent.

- The most common method of correlating the rule consequent with the truth value of the rule antecedent is to cut the consequent membership function at the level of the antecedent truth. This method is called **clipping**. Since the top of the membership function is sliced, the clipped fuzzy set loses some information. However, clipping is still often preferred because it involves less complex and faster mathematics, and generates an aggregated output surface that is easier to defuzzify.

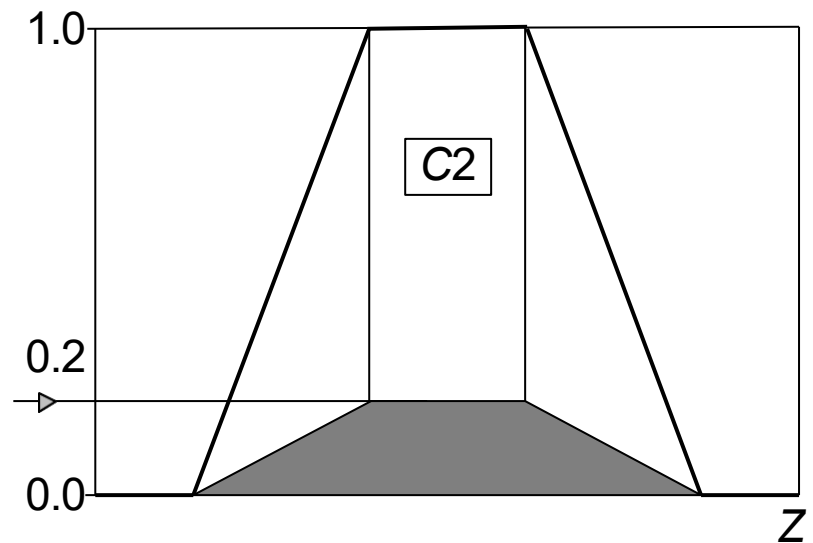
- While clipping is a frequently used method, **scaling** offers a better approach for preserving the original shape of the fuzzy set. The original membership function of the rule consequent is adjusted by multiplying all its membership degrees by the truth value of the rule antecedent. This method, which generally loses less information, can be very useful in fuzzy expert systems.

Clipped and scaled membership functions

*Degree of
Membership*



*Degree of
Membership*

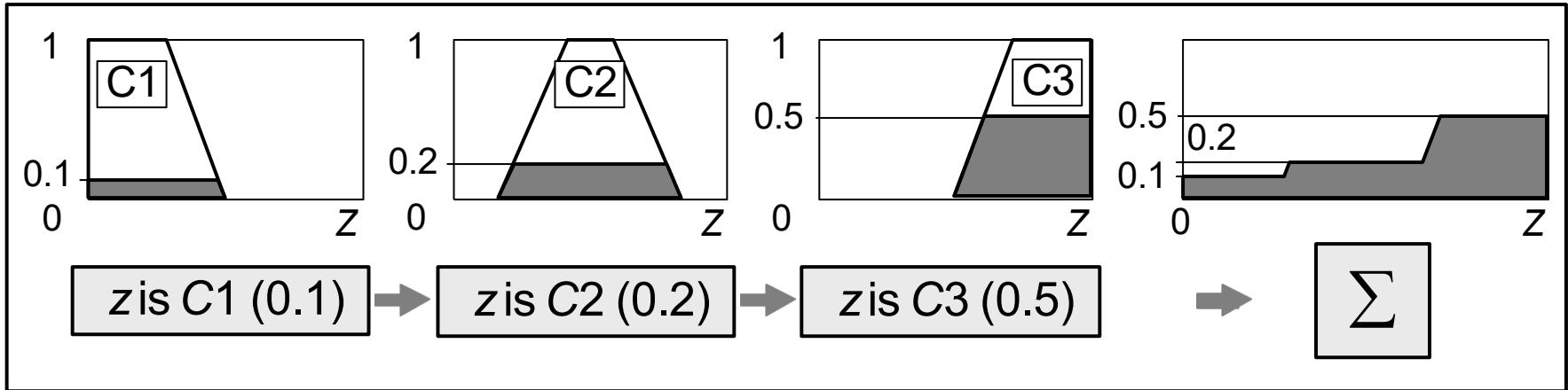


Step 3: Aggregation of the rule outputs

Aggregation is the process of unification of the outputs of all rules. We take the membership functions of all rule consequents previously clipped or scaled and combine them into a single fuzzy set.

The input of the aggregation process is the list of clipped or scaled consequent membership functions, and the output is one fuzzy set for each output variable.

Aggregation of the rule outputs



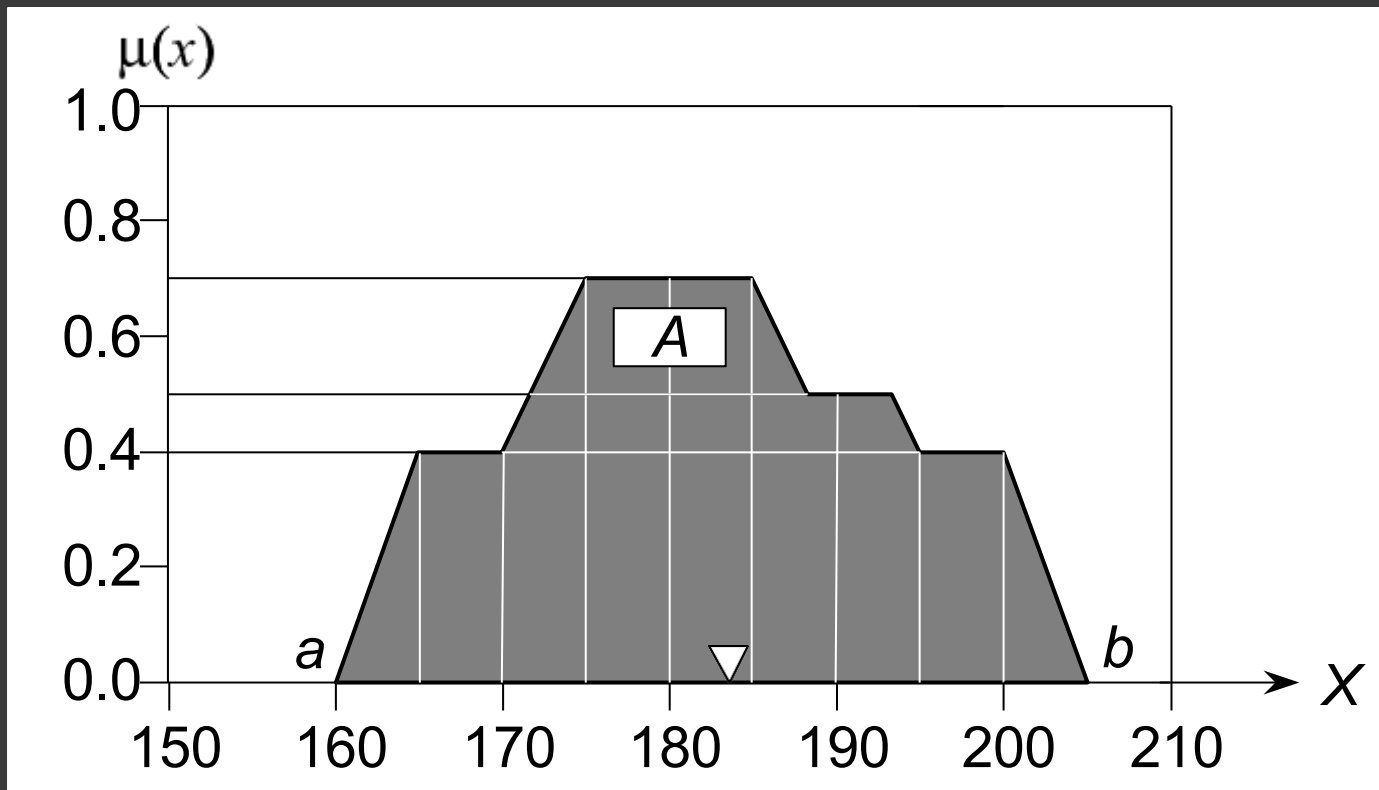
Step 4: Defuzzification

The last step in the fuzzy inference process is defuzzification. Fuzziness helps us to evaluate the rules, but the final output of a fuzzy system has to be a crisp number. The input for the defuzzification process is the aggregate output fuzzy set and the output is a single number.

- There are several defuzzification methods, but probably the most popular one is the **centroid technique**. It finds the point where a vertical line would slice the aggregate set into two equal masses. Mathematically this **centre of gravity (COG)** can be expressed as:

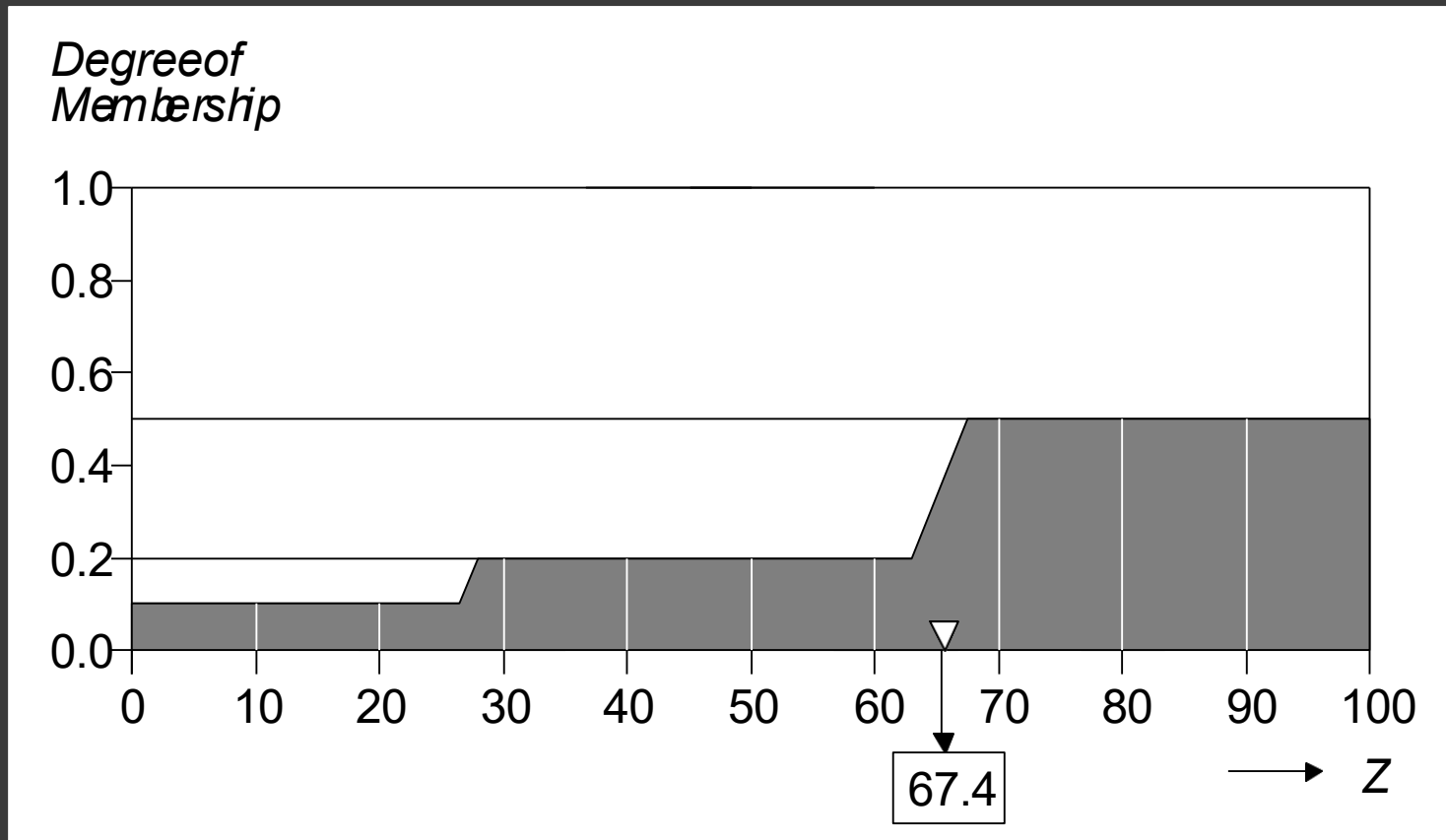
$$COG = \frac{\int_a^b \mu_A(x) x dx}{\int_a^b \mu_A(x) dx}$$

- Centroid defuzzification method finds a point representing the centre of gravity of the fuzzy set, A , on the interval, ab .
- A reasonable estimate can be obtained by calculating it over a sample of points.



Centre of gravity (COG):

$$COG = \frac{(0+10+20) \times 0.1 + (30+40+50+60) \times 0.2 + (70+80+90+100) \times 0.5}{0.1+0.1+0.1+0.2+0.2+0.2+0.2+0.5+0.5+0.5+0.5} = 67.4$$



Sugeno fuzzy inference

- Mamdani-style inference, as we have just seen, requires us to find the centroid of a two-dimensional shape by integrating across a continuously varying function. In general, this process is not computationally efficient.
- **Michio Sugeno** suggested to use a single spike, a *singleton*, as the membership function of the rule consequent. A **singleton**, or more precisely a **fuzzy singleton**, is a fuzzy set with a membership function that is unity at a single particular point on the universe of discourse and zero everywhere else.

Sugeno-style fuzzy inference is very similar to the Mamdani method. Sugeno changed only a rule consequent. Instead of a fuzzy set, he used a mathematical function of the input variable. The format of the **Sugeno-style fuzzy rule** is

IF x is A
AND y is B
THEN z is $f(x, y)$

where x , y and z are linguistic variables; A and B are fuzzy sets on universe of discourses X and Y , respectively; and $f(x, y)$ is a mathematical function.

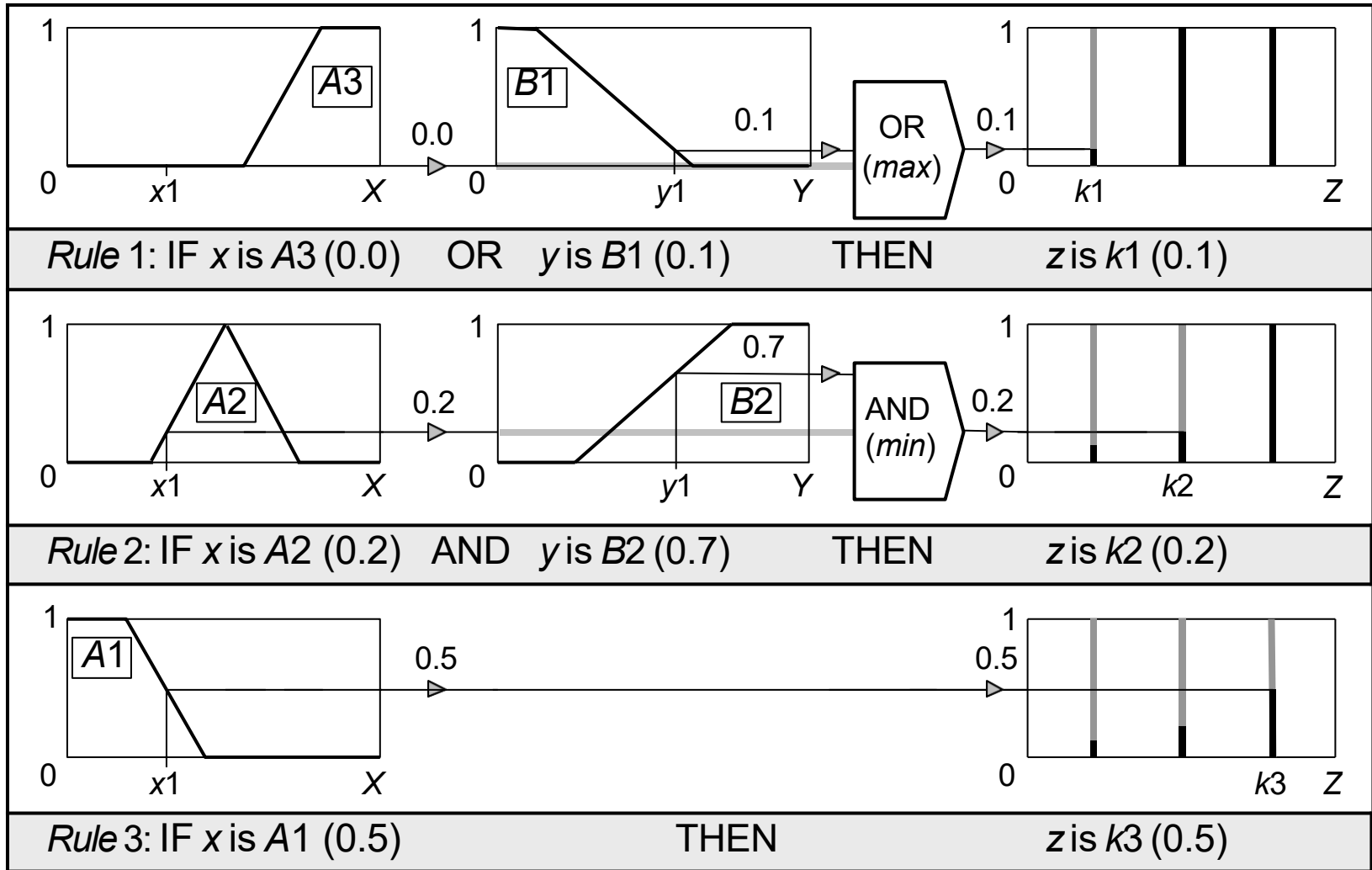
The most commonly used **zero-order Sugeno fuzzy model** applies fuzzy rules in the following form:

IF x is A
AND y is B
THEN z is k

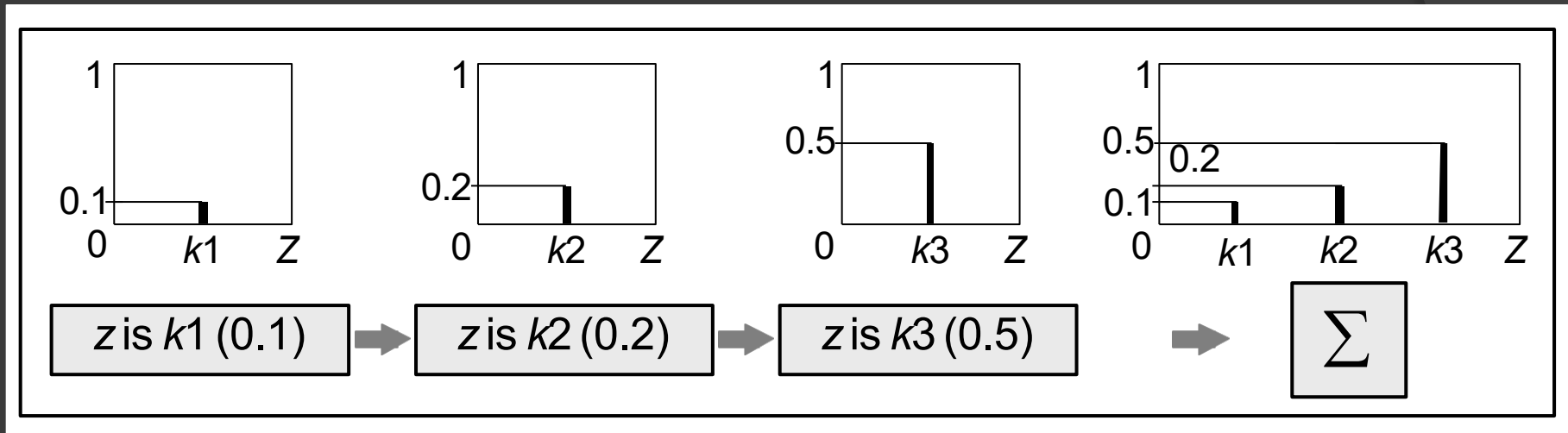
where k is a constant.

In this case, the output of each fuzzy rule is constant. All consequent membership functions are represented by singleton spikes.

Sugeno-style rule evaluation



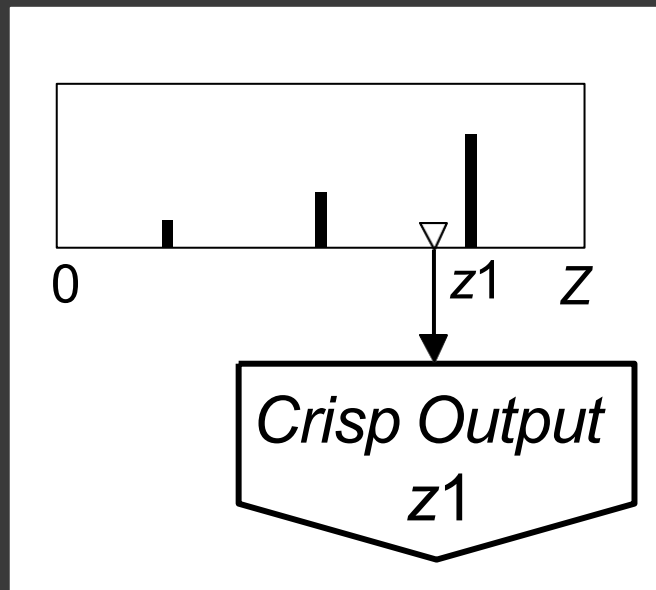
Sugeno-style aggregation of the rule outputs



Weighted average (WA):

$$WA = \frac{\mu(k1) \times k1 + \mu(k2) \times k2 + \mu(k3) \times k3}{\mu(k1) + \mu(k2) + \mu(k3)} = \frac{0.1 \times 20 + 0.2 \times 50 + 0.5 \times 80}{0.1 + 0.2 + 0.5} = 65$$

Sugeno-style defuzzification



How to make a decision on which method to apply – Mamdani or Sugeno?

- Mamdani method is widely accepted for capturing expert knowledge. It allows us to describe the expertise in more intuitive, more human-like manner. However, Mamdani-type fuzzy inference entails a substantial computational burden.
- On the other hand, Sugeno method is computationally effective and works well with optimisation and adaptive techniques, which makes it very attractive in control problems, particularly for dynamic nonlinear systems.

More Examples for Mamdani Fuzzy Models

- **Example #1**

Single input single output Mamdani fuzzy model with 3 rules:

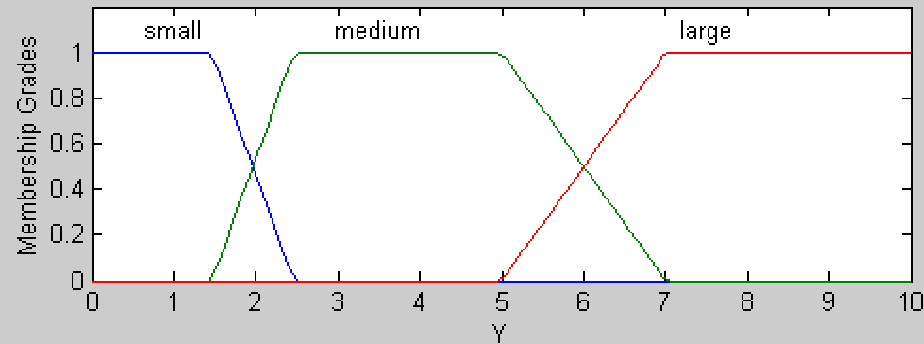
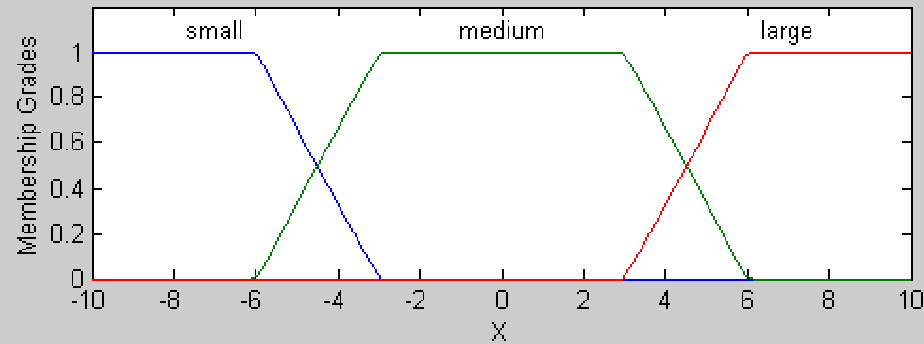
If X is small then Y is small $\rightarrow R_1$

If X is medium then Y is medium $\rightarrow R_2$

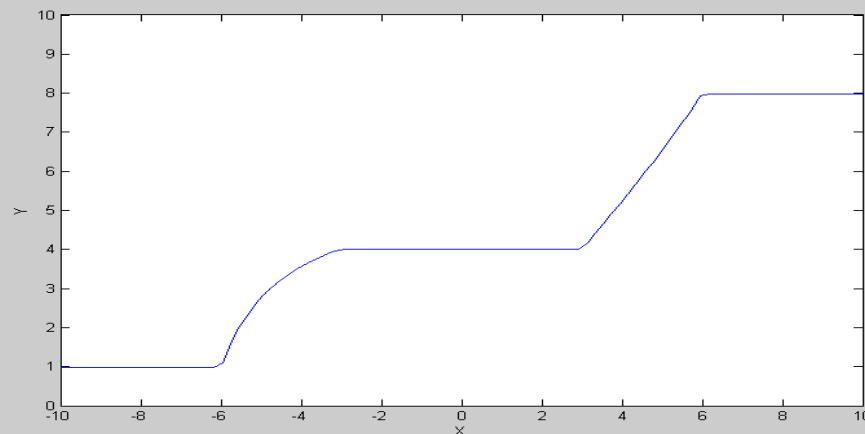
Is X is large then Y is large $\rightarrow R_3$

X = input $\in [-10, 10]$ Y = output $\in [0, 10]$

Using centroid defuzzification, we obtain the following overall input-output curve



Single input single output antecedent & consequent MFs



Overall input-output curve

- **Example #2** (Mamdani Fuzzy models)

Two input single-output Mamdani fuzzy model with 4 rules:

If X is small & Y is small then Z is negative large

If X is small & Y is large then Z is negative small

If X is large & Y is small then Z is positive small

If X is large & Y is large then Z is positive large

More Examples for Sugeno Fuzzy Models

Example 1: Single output-input Sugeno fuzzy model with three rules

If X is small then $Y = 0.1X + 6.4$

If X is medium then $Y = -0.5X + 4$

If X is large then $Y = X - 2$

If “small”, “medium” & “large” are nonfuzzy sets then the overall input-output curve is a piece wise linear

Example 2:

Two-input single output fuzzy model with 4 rules

R_1 : if X is small & Y is small then $z = -x + y + 1$

R_2 : if X is small & Y is large then $z = -y + 3$

R_3 : if X is large & Y is small then $z = -x + 3$

R_4 : if X is large & Y is large then $z = x + y + 2$